

Charlie Marx

CURRICULUM VITAE - DECEMBER 2019

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EDUCATION	Haverford College B.S. in Computer Science (Major GPA: 3.97/4.0) B.S. in Mathematics (Major GPA: 4.0/4.0) Minor in Statistics <i>Relevant Coursework:</i> Machine Learning · Linear Algebra · Algorithms Advanced Topics in Probability · Abstract Algebra · Real Analysis Programming Languages · Multivariable Calculus · Theory of Computation	2016 - MAY 2020 (EXPECTED) GPA: 3.94/4.0
AWARDS & HONORS	CRA Outstanding Undergraduate Researcher Runner-Up One of nine combined runners-up and recipients across North America. Beckman Scholars Award National award for excellence in research including \$26,000 of research and travel funding. Farmers Insurance Standout Student of the Year Regional award for academic and community engagement awarded to one student each year.	2019 2018 2016
PUBLICATIONS	MACHINE LEARNING METHODS 1. Disentangling Influence: Using Disentangled Representations to Audit Model Predictions Charles T. Marx , Richard L. Phillips, Sorelle A. Friedler, Carlos Scheidegger, Suresh Venkatasubramanian <i>NeurIPS – Conference on Neural Information Processing Systems</i> , 2019 2. Predictive Multiplicity in Classification Charles T. Marx , Flavio du Pin Calmon, Berk Ustun <i>NeurIPS Workshop on Human-Centered Machine Learning</i> , 2019 Accepted for Oral Presentation MACHINE LEARNING APPLICATIONS 3. Examining Inputs of Biogenic and Oil-Derived Hydrocarbons in Surface Waters Following the Deepwater Horizon Oil Spill Helen White, Charles T. Marx , David Valentine, Charles Sharpless, Christoph Aeppli, Kelsey Gosselin, Veronika Kivenson, Rachel Liu, Robert Nelson, Sean Sylva, Christopher Reddy <i>ACS Earth and Space Chemistry</i> , 2019 4. Rapid Identification of Deepwater Horizon Oil Residues Using X-Ray Fluorescence Anna Michel, Alexandra E. Morrison, Charles T. Marx , Helen White <i>ACS Environmental Science & Technology Letters</i> , 2018	
RESEARCH EXPERIENCE	Research Assistant in Machine Learning <i>Harvard University</i> Studied predictive multiplicity: the ability for datasets to admit multiple equally accurate models with conflicting predictions. Developed integer programming tools and software to measure the severity of multiplicity. Compiled findings into first-author conference and workshop paper. <i>Advisors:</i> Flavio Calmon and Berk Ustun Research Assistant in Machine Learning <i>Haverford College</i> Designed new methods to quantify indirect feature influence in machine learning models using disentangled representations and adversarial learning. <i>Advisor:</i> Sorelle Friedler Research Assistant in Chemistry <i>Haverford College</i> Designed machine learning tools to optimize model performance, robustness, and interpretability for on-site oil identification. <i>Advisor:</i> Helen White	SUMMER 2019 2017 – PRESENT 2017 – 2018

	Summer Research Intern <i>Prince William Sound Science Center</i> Automated salmon and herring tracking in Alaska using machine learning and sonar data. <i>Advisor:</i> Scott Pegau	SUMMER 2018
	Research Assistant in Computer Vision <i>Haverford College</i> Analyzed 100,000 handwritten texts using recurrent and convolutional neural networks. <i>Advisor:</i> Andrew Janco	2017
ACADEMIC ACTIVITIES	Teaching Assistant in Machine Learning Held weekly office hours and graded for an upper-level machine learning course.	2019 – PRESENT
	Peer Tutor Tutored students in statistics, linear algebra, and machine learning.	2018 – 2019
	Computer Science Faculty Search Committee Student representative on the computer science tenure-track faculty search committee.	2018 – 2019
INVITED & CONTRIBUTED TALKS	Simons Institute for the Theory of Computing <i>Berkeley, CA</i> Disentangling Influence (45-minute talk)	2019
	INFORMS Annual Meeting <i>Seattle, WA</i> Predictive Multiplicity in Classification (20-minute talk)	2019
	FAT* Conference <i>New York, NY</i> Auditing Black Box Models (60-minute tutorial) <i>with Carlos Scheidegger and Suresh Venkatasubramanian</i>	2018
	GOMOSSES Conference <i>New Orleans, LA</i> Examining Inputs of Biogenic and Oil-Derived Hydrocarbons (15-minute talk)	2018
POSTER PRESENTATIONS	NeurIPS Conference <i>Vancouver, BC</i> Disentangling Influence: Using Disentangled Representations to Audit Model Predictions	2019
	NeurIPS Human-Centered Machine Learning Workshop <i>Vancouver, BC</i> On the Multiplicity of Predictions in Machine Learning	2019
	Beckman National Symposium <i>Irvine, CA</i> Indirect Influence in Machine Learning Models	2019
	Beckman Regional Symposium <i>New York, NY</i> Mitigating Social Discrimination in Deep Learning	2018
	KINSC Undergraduate Research Symposium <i>Haverford, PA</i> Examining Biogenic Hydrocarbons After the Deepwater Horizon Oil Spill	2017
FIELDWORK	Alaska Prince William Sound Research Charter	2018
	Gulf of Mexico Shoreline Research Tour	2017
EXTRA- CURRICULAR ACTIVITIES	Captain of Haverford College Men’s Club Soccer	2018 – 2019
	Leader of Haverford College Outdoors Club	2017 – 2018
	President of Furniture for Newly Housed Families Organization	2015 – 2017
SKILLS & INTERESTS	Languages: <i>Fluent:</i> Python, R, LaTeX · <i>Familiar:</i> SQL, C, Unix Tools: Keras/Tensorflow, PyTorch, Pandas, CPLEX Interests: Backpacking, Gardening, Soccer, Folk Music, Cooking	